

LAB EXPERIMENT RECORD

Set 11: Product Category Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

AIM

To analyze product category sales using a funnel chart, table, pie chart, and Tableau dashboard for interactive exploration of product category data.

PROGRAM 1: FUNNEL CHART OF PRODUCT CATEGORY SALES

Aim

To generate a funnel chart to analyze the sales conversion process for each product category.

R Code

R

```

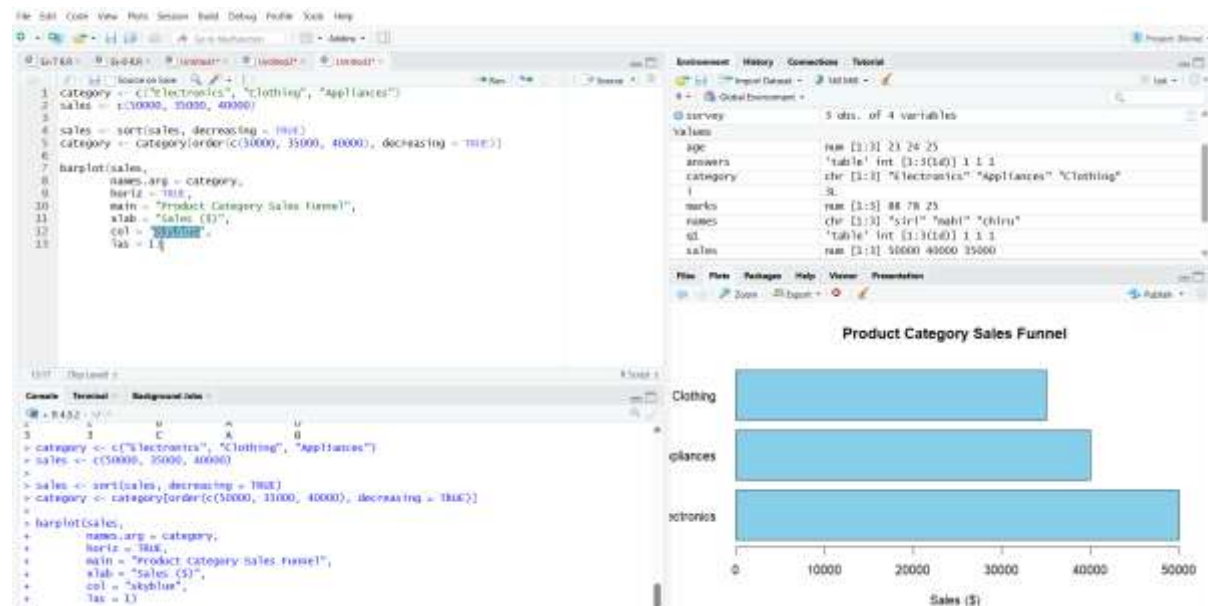
category <- c("Electronics", "Clothing", "Appliances")
sales <- c(50000, 35000, 40000)
sales <- sort(sales, decreasing = TRUE)
category <- category[order(c(50000, 35000, 40000), decreasing = TRUE)]
barplot(sales,
        names.arg = category,
        horiz = TRUE,
        main = "Product Category Sales Funnel",
        xlab = "Sales ($)",
        col = "skyblue",
        las = 1)

```

Output

A horizontal funnel-style chart showing the sales values of Electronics, Appliances, and Clothing.

Observation



Electronics has the highest sales of \$50,000, followed by Appliances with \$40,000 and Clothing with \$35,000.

PROGRAM 2: TABLE OF PRODUCT CATEGORY SALES

Aim

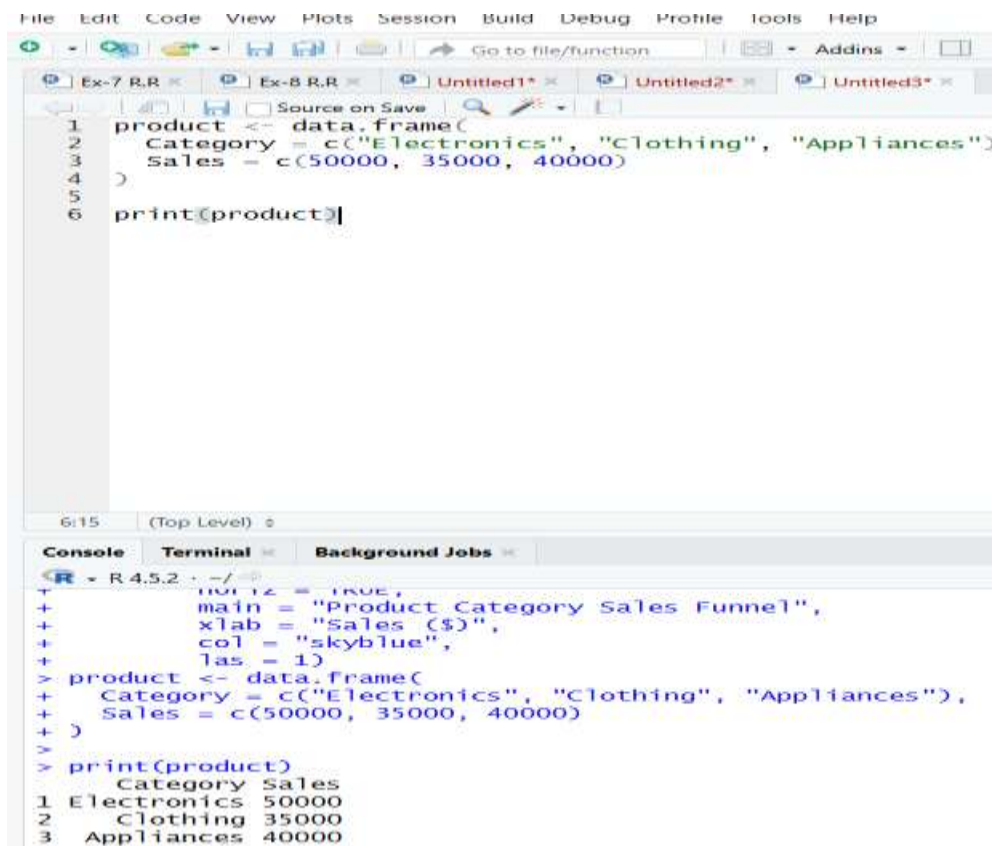
To build a table displaying the sales data for each product category.

R Code

R

```
product <- data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
print(product)
```

Output



The screenshot shows the R Studio interface. The script editor at the top contains the following R code:

```
1 product <- data.frame(  
2   Category = c("Electronics", "Clothing", "Appliances")  
3   Sales = c(50000, 35000, 40000)  
4 )  
5  
6 print(product)
```

The console at the bottom shows the execution of the code, resulting in the following output:

```
> product <- data.frame(  
+   Category = c("Electronics", "Clothing", "Appliances"),  
+   Sales = c(50000, 35000, 40000)  
+ )  
>  
> print(product)  
  Category Sales  
1 Electronics 50000  
2  Clothing 35000  
3 Appliances 40000
```

Observation

The table clearly displays the sales amount for each product category. Electronics has the highest sales, while Clothing has the lowest sales.

PROGRAM 3: TABLEAU DASHBOARD

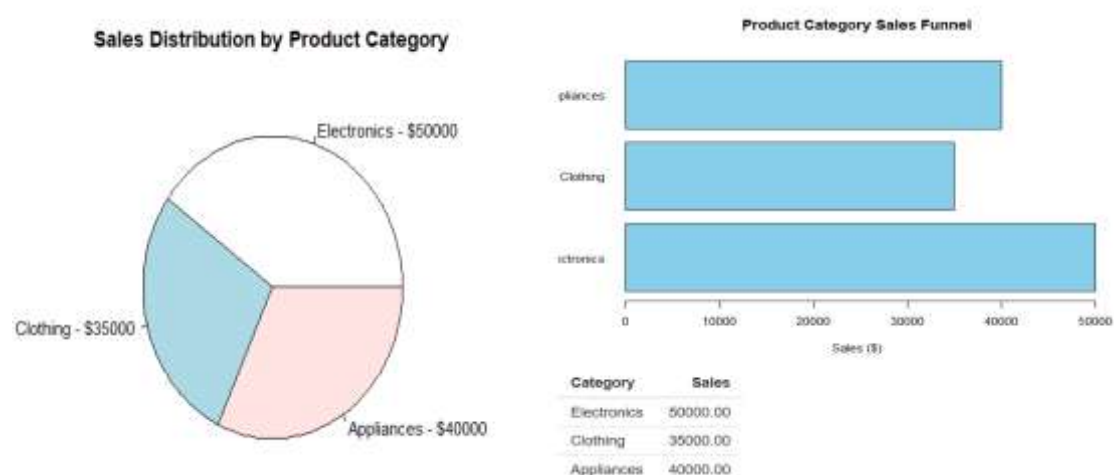
Aim

To develop a Tableau dashboard combining the pie chart, funnel chart, and table for interactive exploration of product category data.

Steps

1. Open Tableau.
2. Import the Product Categories dataset.
3. Create a worksheet for the **Pie Chart**.
4. Create another worksheet for the **Funnel Chart**.
5. Create a third worksheet for the **Sales Table**.
6. Select **Dashboard** → **New Dashboard**.
7. Drag the pie chart into the dashboard.
8. Drag the funnel chart into the dashboard.
9. Drag the sales table into the dashboard.
10. Arrange the charts and table properly.
11. Add suitable titles and labels.
12. Add filters if required for interactive exploration.
13. Save the Tableau dashboard.

Output



An interactive Tableau dashboard containing the pie chart, funnel chart, and sales table.

Observation

The Tableau dashboard provides an interactive view of product category sales. It helps users compare sales values and understand the contribution of each category.

PROGRAM 4: PIE CHART OF SALES DISTRIBUTION

Aim

To create a pie chart representing the distribution of sales across product categories with labels.

R Code

R

```
category <- c("Electronics", "Clothing", "Appliances")
```

```
sales <- c(50000, 35000, 40000)
```

```
labels <- paste(category, sales, sep = " - $")
```

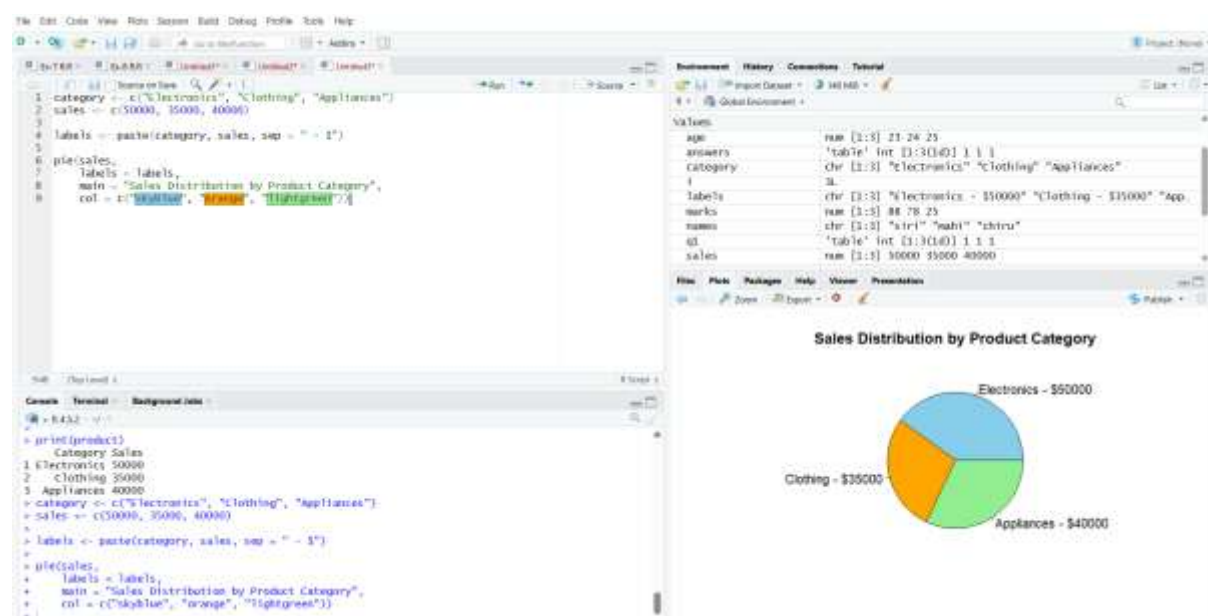
```
pie(sales,
```

```
  labels = labels,
```

```
  main = "Sales Distribution by Product Category",
```

```
  col = c("skyblue", "orange", "lightgreen"))
```

Output



A pie chart showing the sales distribution of Electronics, Clothing, and Appliances with labels.

Observation

The pie chart shows that Electronics contributes the largest share of total sales, followed by Appliances and Clothing.